

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Triangles

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

In triangle ABC, DE is parallel to BC. If $AD=3$, $DB=2$ and $AE=4.5$, find EC.

Answer:

By BPT, $AD/DB = AE/EC$. Thus $3/2 = 4.5/EC$, so $EC = 3$ units.

Question 2

In triangle ABC, DE is parallel to BC. If $AD=4$, $DB=3$ and $AE=6$, find EC.

Answer:

By BPT, $AD/DB = AE/EC$. Thus $4/3 = 6/EC$, so $EC = 4.5$ units.

Question 3

In triangle ABC, DE is parallel to BC. If $AD=5$, $DB=2$ and $AE=7.5$, find EC.

Answer:

By BPT, $AD/DB = AE/EC$. Thus $5/2 = 7.5/EC$, so $EC = 3$ units.

Question 4

In triangle ABC, DE is parallel to BC. If $AD=6$, $DB=4$ and $AE=9$, find EC.

Answer:

By BPT, $AD/DB = AE/EC$. Thus $6/4 = 9/EC$, so $EC = 6$ units.

Question 5

In triangle ABC, DE is parallel to BC. If $AD=2$, $DB=5$ and $AE=3$, find EC.

Answer:

By BPT, $AD/DB = AE/EC$. Thus $2/5 = 3/EC$, so $EC = 7.5$ units.

Question 6

In triangle ABC, DE is parallel to BC. If $AD=7$, $DB=3$ and $AE=10.5$, find EC.

Answer:

By BPT, $AD/DB = AE/EC$. Thus $7/3 = 10.5/EC$, so $EC = 4.5$ units.

Question 7

In triangle ABC, DE is parallel to BC. If $AD=8$, $DB=2$ and $AE=12$, find EC.

Answer:

By BPT, $AD/DB = AE/EC$. Thus $8/2 = 12/EC$, so $EC = 3$ units.

Question 8

In triangle ABC, DE is parallel to BC. If $AD=9$, $DB=6$ and $AE=13.5$, find EC.

Answer:

By BPT, $AD/DB = AE/EC$. Thus $9/6 = 13.5/EC$, so $EC = 9$ units.

Triangles - Continued

Question 9

Two similar triangles have corresponding side ratio 3:5. If the smaller corresponding side is 6 cm, find the larger side.

Answer:

Scale factor = $\frac{5}{3}$. Larger side = $6 \times \frac{5}{3} = 10$ cm.

Question 10

Two similar triangles have corresponding side ratio 4:7. If the smaller corresponding side is 8 cm, find the larger side.

Answer:

Scale factor = $\frac{7}{4}$. Larger side = $8 \times \frac{7}{4} = 14$ cm.

Question 11

Two similar triangles have corresponding side ratio 2:3. If the smaller corresponding side is 9 cm, find the larger side.

Answer:

Scale factor = $\frac{3}{2}$. Larger side = $9 \times \frac{3}{2} = 13.5$ cm.

Question 12

Two similar triangles have corresponding side ratio 5:8. If the smaller corresponding side is 10 cm, find the larger side.

Answer:

Scale factor = $\frac{8}{5}$. Larger side = $10 \times \frac{8}{5} = 16$ cm.

Question 13

Two similar triangles have corresponding side ratio 6:9. If the smaller corresponding side is 12 cm, find the larger side.

Answer:

Scale factor = $\frac{9}{6}$. Larger side = $12 \times \frac{9}{6} = 18$ cm.

Question 14

Two similar triangles have corresponding side ratio 7:11. If the smaller corresponding side is 14 cm, find the larger side.

Answer:

Scale factor = $\frac{11}{7}$. Larger side = $14 \times \frac{11}{7} = 22$ cm.

Question 15

A 1.5 m pole casts a 2 m shadow. A tree casts a 8 m shadow at the same time. Find tree height.

Answer:

By similarity, height/shadow is constant. Tree height = $1.5 \times \frac{8}{2} = 6$ m.

Question 16

A 2 m pole casts a 3 m shadow. A tree casts a 12 m shadow at the same time. Find tree height.

Answer:

By similarity, height/shadow is constant. Tree height = $2 \times \frac{12}{3} = 8$ m.

Triangles - Continued

Question 17

A 1.2 m pole casts a 1.5 m shadow. A tree casts a 10 m shadow at the same time. Find tree height.

Answer:

By similarity, height/shadow is constant. Tree height = $1.2 \times 10/1.5 = 8$ m.

Question 18

A 2.5 m pole casts a 4 m shadow. A tree casts a 20 m shadow at the same time. Find tree height.

Answer:

By similarity, height/shadow is constant. Tree height = $2.5 \times 20/4 = 12.5$ m.

Question 19

A 1.8 m pole casts a 3 m shadow. A tree casts a 15 m shadow at the same time. Find tree height.

Answer:

By similarity, height/shadow is constant. Tree height = $1.8 \times 15/3 = 9$ m.

Question 20

A 3 m pole casts a 5 m shadow. A tree casts a 25 m shadow at the same time. Find tree height.

Answer:

By similarity, height/shadow is constant. Tree height = $3 \times 25/5 = 15$ m.

Question 21

State the main formula, rule or theorem used in Triangles.

Answer:

If a line is parallel to one side of a triangle, it divides the other two sides proportionally (BPT).

Question 22

What do the important symbols or quantities in Triangles represent?

Answer:

Similar triangles have equal corresponding angles and proportional corresponding sides.

Question 23

Write the correct method for solving a Triangles numerical.

Answer:

State the similarity criterion or BPT, write the proportion, then solve for the unknown side.

Question 24

Which unit, sign convention or condition must be checked in a Triangles answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Triangles be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Triangles - Continued

Question 26

What is one common error in Triangles questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Triangles numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Triangles?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Question 29

How can a student decide which formula to use in Triangles?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Triangles?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).